The “Tortoise” Glider

MAE 3050 Final Report

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Introduction

The purpose of this report is for the in class project for MAE 3050—*Introduction to Aeronautics*, where each group is tasked to design a glider to travel as close to 20m as possible from a 5m launch height. The glider will be given an initial velocity from a rubber band launch station that resembles a simple spring mass system. The purpose for this project is to demonstrate the knowledge taught in this course by applying it through the design of a glider to achieve the provided objective.

The design for our glider is based on the principle that we would rather fly slower and longer to reach the 20m goal, rather than try to dart to the finish line. The latter was the thought process behind our first design that was not successful, and this switch in design was also reinforced by the success of other groups during their first trials, where larger but slower flying gliders tended to be more successful. Thus, our final design reflects a larger chord, wingspan, and planform area that allows the glider to reach the 20m range with a high endurance. See Figure 2 for images of the first design and current design side-by-side.

In this report, we will begin with an overview of the glider and the strategy behind its design, the associated assumptions, and calculations. A summary of the technical information will then be provided. Finally, the full design will be unveiled, before providing the bill of materials and an overview of the fabrication.

The name of the glider is based on the tale from *Aesop's Fables* of the race between the tortoise and the hare. The moral of the fable is that "slow and steady wins the race." Our design aims to emulate the principle and, thus, was name "Tortoise."

Glider Description

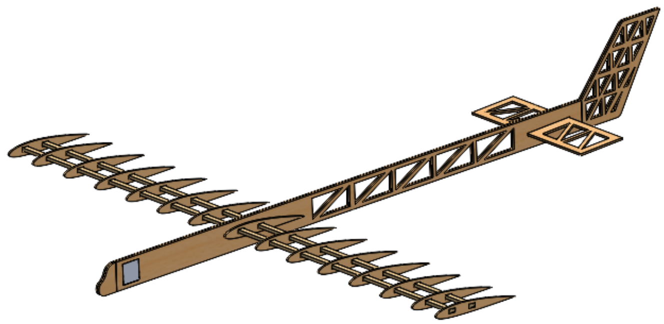

Figure 1: CAD model of redesigned glider

Strategy

The strategy we chose as stated above to achieve this task is to design our glider to have a slower cruising speed; thus we have chosen an anticipated design for the following target parameter inputs:

- Cruise speed: 4 m/s
- Characteristic Chord Length: 0.15 m
- Mass: 0.1 kg
- C_L : 0.75

Assumptions

The assumptions we used are standard atmospheric temperature, pressure, density, and inviscid and incompressible flow approximations. These assumptions and targeted parameters for our design allow us to calculate the remaining characteristics of the glider and allow us to determine if the targeted input parameters are met.

Airfoil Choice

We chose the NACA 2412 airfoil as it provides great performance characteristics for a glider, being cambered it has a high C_L value for small angles of attack, and low C_D for those respective angles. This allows for sufficient C_L/C_D approximations. Secondly, it has a sleek, slim, and gently cambered airfoil design. This allows our glider to be aerodynamic and efficient just from visual identification, while its performance reinforces this from its charts on airfoil tools website. The chosen angle of attack is 5°. This was chosen to yield a C_L value of 0.75, which is close to the $(C_L/C_D)_{\max}$ for this airfoil, but based on the plot however for this airfoil that occurs too close to the stall angle of attack for our comfort at 7°; thus we preferred to choose a more conservative value of 5°. Despite this, our C_L/C_D value is 21.43, which is sufficient for a glider.

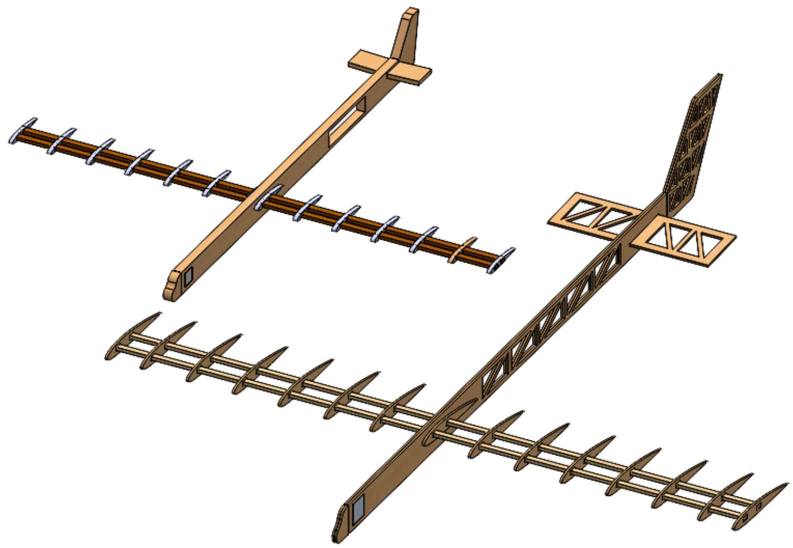

Figure 2: Side-by-side isometric view of first (upper-left) and current (bottom-right) glider designs

Changes due to Test Flight Learning

Our test flight with the first iteration of our glider was very interesting but overall did not perform well. Our glider only traveled a short distance and quickly pitched up after leaving the launching block. We could also see that our speed was pretty high relative to the challenge goals and we wanted to decrease our speed going into the second iteration, and thus we changed our speed from 7 m/s in the first iteration to roughly 4 m/s in the second iteration, and this will also bring us closer to our ideal of having a lower Reynolds number and puts us at around 40000 Re. To fix our pitching up we are going to make sure that we have a better balancing methodology for our second glider and will make sure to balance it better using the provided clay, and we will also be adding cutouts across the fuselage and rear stabilizers to ensure that our glider's center of gravity is roughly where the wings are. We weren't able to glean too much information about our glider's build quality since it fell early into our flight but it seemed good. We also noticed that gliders with bigger wings tended to perform better, so we will be aiming to do that in our final/second iteration as well.

Our observations as outlined are quantified by the following changes. A decrease in total flight speed from 7 m/s to 4 m/s will align the design with the goals of the challenge. We are increasing our characteristic chord length to 15 cm to result in larger wings and an increased lift. We are also increasing our angle of attack to improve our C_L to further increase our lift as well. We will be both checking our center of gravity not only in our CAD but also in real life once we add our clay and that will allow us to more accurately ensure our glider doesn't pitch up or down during flight. We also had a hard time building our glider due to the lack of tolerancing and that resulted in a lot of sanding while we built which reduced the precision of our build, thus we will now add a 1/16" kerf to allow for the tolerances due to laser cutting. We will also be increasing our mass budget since we now have larger wings and larger glider overall. We are also removing the dihedral since it added extra complexity that we weren't able to replicate in real life as we built our glider the first time.

Calculations and Choices

Velocity and Reynolds Number

Based on our experiences with our test flight we decided to change our target velocity from 7 m/s in our first glider to 4 m/s in this iteration, this also kept our Reynolds number roughly the same which allows us to keep the same regime in our C_L approximations.

We can calculate our speed of sound and our mach number as following.

$$a = \sqrt{\gamma * R * T} = \sqrt{1.4 * 287 * 293}$$
$$\text{Where Room Temp is 293K}$$
$$\text{Thus } a = 337.208 \text{ m/s}$$

$$\text{Mach Number } M = \frac{V}{a} = \frac{4}{337.208} = 0.011$$

We chose our characteristic chord to be 15 cm which is larger than the 10 cm in our first iteration because we realized our chord was too small to properly build and it also underutilized the material we had. This time we are using a longer chord to utilize more material and increase the overall surface area and lift we can get out of our wings. Based on the characteristic chord we can find our characteristic Reynolds number.

$$Re = \frac{\rho * V * c}{\mu} = \frac{1.225 * 4 * 0.15}{1.81E - 5}$$
$$Re = 40,608$$

This Reynolds number is within the laminar flow regime as we want, and the overall flight time will be roughly 5 seconds.

C_L, C_D, and Angle of Attack

Earlier we found that we wanted a higher coefficient of lift compared to our earlier glider set up since we wanted to have more lift on a larger area and thus that means that we now want a target C_L of 0.75. Based on our graphs for our airfoil we can see that this amount of lift happens at an angle of attack of 5 degrees. We also found that this corresponds to a C_D of 0.035 based on the graph as well. Since our airfoil has the coefficient of lift almost 0 at 0 degrees angle of attack we can find that our $\frac{C_L}{\alpha}$ is as follows.

$$\frac{C_L}{\alpha} = \frac{C_L}{\alpha} = \frac{0.75}{5} = 0.15$$

Wing Dimensions and Glider Mass

We found that our mass will be a bit higher in this iteration and be around 100g since our last iteration with smaller wings was 90g and even though we are adding a lot of mass reductions in our fuselage, we still believe the larger wings will make it slightly heavier. This means that our lift will be as follows.

$$\text{Lift} = m * g = 0.1 * 9.81 = 0.981 \text{ N}$$

Next we can find that surface area that we need using the lift equation and isolating S from that equation. Our new surface area is roughly 3 times our first iteration.

$$L = 0.5 * C_L * \rho * V^2 * S$$

$$S = \frac{L}{0.5 * C_L * \rho * V^2}$$

Then by substituting values we can find that:

$$S = \frac{0.981}{0.5 * 0.75 * 1.225 * 4^2} = 0.133 \text{ m}^2$$

We are choosing to use an AR of 6 for the main wing compared to the AR of 10 we used in the first iteration because the lower aspect ratio reduces the span while still increasing the total wing area by increasing the chord length such that we can use two spars to support the total wingspan, helping with the structural integrity and shaping of the wing surface. We also chose to not have a taper ratio and not have a dihedral angle since we felt that due to the hand-built nature of our glider it would add a lot of complexity that we don't know we could achieve through our building techniques, and even if we could achieve it, we don't think it would add much benefit. Based on these values and our characteristic chord length of 15 centimeters we can find the following for our wing dimensions.

$$\text{AR} = \frac{b^2}{S} = 6$$

$$b = \sqrt{\text{AR} * S} = \sqrt{6 * 0.133} = 0.894 \text{ m}$$

$$\text{AR} = \frac{b}{c}$$

$$c = \frac{b}{\text{AR}} = \frac{0.894}{6} = 0.149 \text{ m}$$

Then we can recalculate our Reynolds number to get a value accurate to the wing and we can find that the new Reynolds number is as follows.

$$Re = \frac{\rho * V * c}{\mu} = \frac{1.225 * 4 * 0.149}{1.81E - 5}$$

$$Re = 40,377$$

Tail Dimensions

Next for the tail we decided to increase its size slightly from the first iteration to ensure that our stability constraints can still be met. We chose to have our horizontal stabilizer have a moment arm of 55 cm, and our vertical stabilizer have a moment arm of 60 cm. Along with this we decided to keep our coefficients the same for both stabilizers since they are standard for gliders, and we chose a coefficient of 0.5 for our horizontal stabilizer, and a coefficient of 0.05 for our vertical stabilizer. Using these values we can use the equations used to find these coefficients and isolate to find the surface area needed for each stabilizer. HT is used to denote the horizontal stabilizer, VT for the vertical stabilizer, and w for the main wing.

Horizontal Stabilizer

$$C_{HT} = \frac{L_{HT} * S_{HT}}{c_w * S_w}$$
$$S_{HT} = \frac{C_{HT} * c_w * S_w}{L_{HT}}$$
$$S_{HT} = \frac{0.5 * 0.149 * 0.133}{0.55} = 0.0181 \text{ m}^2$$

Vertical Stabilizer

$$C_{VT} = \frac{L_{VT} * S_{VT}}{b_w * S_w}$$
$$S_{VT} = \frac{C_{VT} * b_w * S_w}{L_{VT}}$$
$$S_{VT} = \frac{0.05 * 0.894 * 0.133}{0.55} = 0.010 \text{ m}^2$$

Then to find the dimensions of the tails we are using an AR of 3 for the horizontal stabilizer and a AR of 2 for the vertical stabilizer since that is the common value we found for gliders, and we are not changing this from our first iteration. Then using a similar analysis to what we did for our main wing we can find the following values for the sizing of the horizontal and vertical stabilizers.

Horizontal Stabilizer

$$\text{AR}_{HT} = \frac{b_{HT}^2}{S_{HT}} = 3$$
$$b_{HT} = \sqrt{\text{AR}_{HT} * S_{HT}} = \sqrt{3 * 0.0181} = 0.233 \text{ m}$$
$$\text{AR}_{HT} = \frac{b_{HT}}{c_{HT}}$$
$$c_{HT} = \frac{b_{HT}}{\text{AR}_{HT}} = \frac{0.233}{3} = 0.077 \text{ m}$$

Vertical Stabilizer

$$\text{AR}_{VT} = \frac{b_{VT}^2}{S_{VT}} = 2$$
$$b_{VT} = \sqrt{\text{AR}_{VT} * S_{VT}} = \sqrt{2 * 0.01} = 0.141 \text{ m}$$
$$\text{AR}_{VT} = \frac{b_{VT}}{c_{VT}}$$
$$c_{VT} = \frac{b_{VT}}{\text{AR}_{VT}} = \frac{0.141}{2} = 0.0705 \text{ m}$$

Static Stability

The system of equations below describes the quarter-chord location of the wing ℓ_W and the quarter-chord location of the horizontal stabilizer ℓ_E with respect to the center of aerodynamic force.

$$\frac{\ell_W}{\ell_E} = -\frac{S_E}{S_W}\frac{1 - \frac{2}{\text{AR}_W}}{1 - \frac{2}{\text{AR}_E}}$$
$$d = \ell_W + \ell_E$$

The horizontal stabilizer's quarter-chord is located at a distance of $d = L_{HT} = 0.55$ m from the quarter-chord of the wing. The values of S_W, S_E, AR_W, and AR_E are given in the Glider Characteristics table. We can set up the system of equations as a matrix equation and then solve in MATLAB for ℓ_W:

$$\begin{bmatrix} 1 & -\frac{S_E}{S_W}\frac{1-\frac{2}{\mathrm{AR}_W}}{1-\frac{2}{\mathrm{AR}_E}} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \ell_W \\ \ell_S \end{bmatrix} = \begin{bmatrix} 0 \\ d \end{bmatrix}$$

Using `MATLAB`, we get $\ell_W = 0.155$ m $= 6.102$ in; the center of aerodynamic force $X_A = -6.102$ in where positive x is towards the nose. From the CAD, we are given that the center of mass is located 5.37 in aft of the leading edge of the wing or 3.889 in aft of the quarter-chord for a chord of $c_W = 5.905$ in; $X_G = -3.889$ in. Thus, the static margin of an aircraft is

$$\text{static margin} = -\frac{X_A - X_G}{\bar{c}_W} = -\frac{(-6.102) - (-3.889)}{5.906} = 0.375$$

An aircraft is considered statically stable if the static margin is greater than 0.1. **This design is statically stable.**

Pull Back Distance

For the pull back distance we can use an energy equality between the spring potential energy of the rubber band to the kinetic energy of the glider. We are told that the K of the rubber band is 61 N/m, and the mass of our glider is 100 grams, and the velocity that we want is 4 m/s^2. Thus we can find the following.

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$
$$mv^2 = kx^2$$
$$x = \sqrt{\frac{mv^2}{k}}$$
$$x = \sqrt{\frac{0.1 * 4^2}{61}} = 0.16m$$

We are rounding to 18 cm to account for friction etc.

Drag and Sink Rate

Based on our earlier choices and findings we can find our coefficient of induced drag and our coefficient of skin friction drag using the following methods.

$$C_{di} = \frac{C_L^2}{\pi * AR} = \frac{0.75^2}{\pi * 6} = 0.029$$
$$C_{df} = \frac{1.328}{\sqrt{\mathrm{Re}}} = \frac{1.328}{\sqrt{40376}} = 0.006$$

This also matches well with the value we received from our airfoil of C_D equals 0.035. From our airfoil we can also find that our max coefficient of lift is around 1.2. Using this we can find our stall speed.

$$\text{Stall Speed} = \sqrt{\frac{2 * Lift}{\rho * S_w * C_{L,max}}} = \sqrt{\frac{2 * 0.981}{1.225 * 0.133 * 1.2}} = 3.16m/s$$

This means that we are above the stall speed in our configuration and 4 m/s is well above the stall speed and thus with this speed we should be able to fly well.

Next we can find our glide ratio. Here we have our D as our distance off the ground which is 5 meters, and L is our distance to travel which is 20 meters.

$$\text{Glide Ratio is } \tan(\gamma) = \frac{-D}{L} = \frac{-5}{20} = 0.25$$
$$\gamma = tan^{-1}(0.25) = -0.244 \text{ rad}$$

Then we can find our sink rate and our time to our target length of 20 meters is represented as t which is 5 seconds.

$$\text{Sink Rate is } V_s = \frac{D}{t} = \frac{5}{5} = 1m/s$$

Using this we can find our ideal cruise speed to ensure that we do not sink too fast, this can be calculated by isolating V from the sink rate equation.

$$V_s = -V * sin(\gamma)$$
$$V = \frac{-V_s}{sin(\gamma)} = \frac{-1}{sin(-0.244)} = 4.08m/s$$

This is very close to our existing speed of 4 m/s which means that our cruise speed of 4 m/s is optimal for the glider.

Technical Summary

See the previous sections for the source of the values below.

Quantity	**Symbol**	**Value**	**Unit**
Chord Length	c_W	0.149	m
Aspect Ratio	AR_W	6	–
Wing Span	b_W	0.895	m
Planform Area	S_W	0.13347	m^2
Volume Coeff. Horizontal Tail	C_{HT}	0.5	–
Volume Coeff. Vertical Tail	C_{VT}	0.05	–
Moment Arm Horizontal Tail	L_{HT}	0.55	m
Moment Arm Vertical Stabilizer	L_{VT}	0.6	m
Lift Coefficient	C_L	0.75	–
Drag Coefficient	C_D	0.035	–
Induced Drag Coeff.	C_{Di}	0.02984	–
Friction Drag Coeff.	C_{Df}	0.00661	–
Angle of Attack	α	5	$^\circ$
Flight Time	t	5	s
Flight Speed	V	4	m/s
Glide Ratio	γ	-0.245	–
Sink Rate	V_{sink}	-1	m/s
Reynolds Number	Re	40,377	–
Mach Number	M	0.01186	–
Pull Back Distance	d_{pull}	0.162	m
Weight	W	0.981	N
Stall Speed	V_{stall}	3.16	m/s

Table 1: Glider Geometric and Aerodynamic Characteristics

Design & Bill of Materials

CAD Model Images

Figures 3–5 show the model of the glider as created in SOLIDWORKS.

The ACS is the "aircraft coordinate system." The origin is located at the wing quarter-chord. $+x$ points from tail to nose. $+y$ points to the right wingtip. $+z$ points down. The source of this coordinate system is the NASA Glenn Research Center. The positive x, y, and z axes correspond to positive pitch (up), roll (right), and yaw (right). This is not the same as the native CAD coordinate system which has its origin passing through the wing leading edge at the root.

The airfoil curve was generated by scaling a unit length NACA 2412 `.csv` file to be 0.15 m. 101 points are used to define the curve.

The tissue paper is not depicted in the CAD. The wing surface and the mass-reduction cut-outs in the fuselage, horizontal, and vertical stabilizer will all be covered with tissue paper.

The mass reduction cutouts both bring the design close to the originally assumed 100 gram estimate for the mass, but also helps in pushing the center of mass forward. The mass based on the CAD model is 112.74 grams (excluding tissue paper, glue, and other bonding agents).

All holes and slots into which another component must fit are over-sized by 1/16th of an inch. During the previous construction, there were significant issues with assembly due to tight-fitting parts.

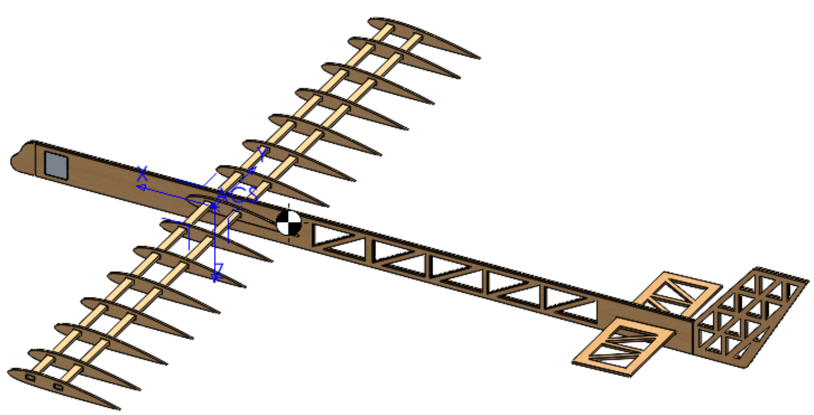

Figure 3: Isometric CAD view with center of mass marker

Bill of Materials

Table 2 prescribes the necessary quantities of all the components in the glider assembly, and prescribes their cost in terms of the *surface area* of the raw material used. This table constitutes the component-wise bill of materials.

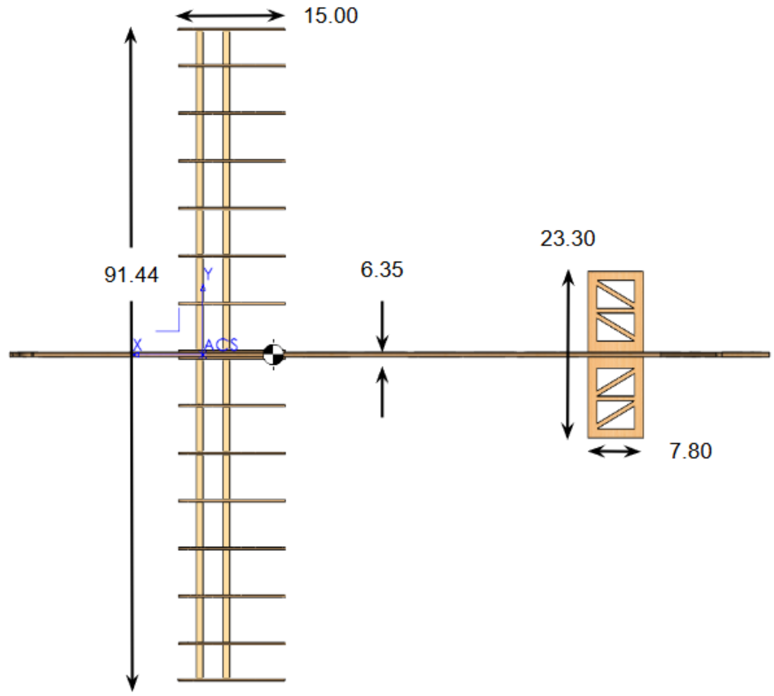

Figure 4: Top View of CAD, dimensioned (centimeters)

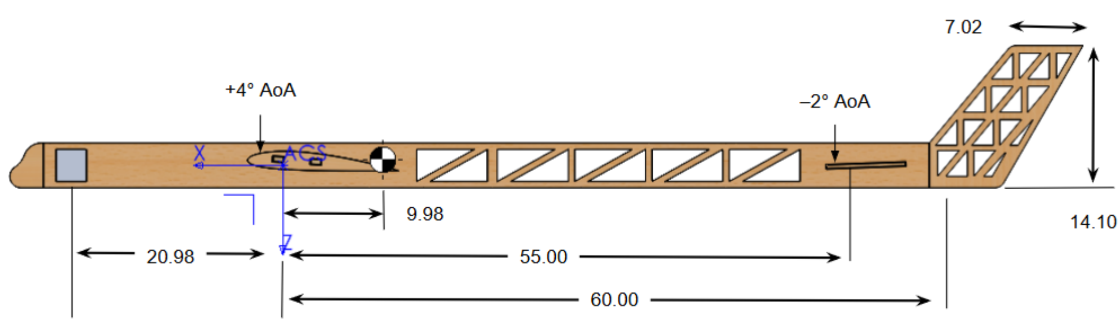

Figure 5: Left View of CAD, dimensioned (centimeters). Distances from aerodynamic surfaces, i.e., wing, horizontal/vertical stabilizer, are measured from their respective quarter-chords.

Component	Material	Cost (in^2)	Qty	Total Area (in^2)
Fuselage	Balsa (large)	60.5	2	121.0
Wing Ribs	Balsa (large)	4.2	20	84.0
Spar	Spruce	13.5	2	27.0
Horizontal Stabilizer	Balsa (small)	28.2	1	28.2
Vertical Stabilizer	Balsa (small)	33.0	2	66.0
Nose	Balsa (small)	2.3	2	4.6
Ballast	Clay	N/A	16 g	N/A
Wing Skin	Tissue Paper	403	1	403.0
Fuselage Skin	Tissue Paper	37.4	1	37.4
HT Skin	Tissue Paper	34.9	1	34.9
VT Skin	Tissue Paper	36.7	1	36.7
Total				**842.8**

Table 2: Bill of Materials (BOM)

Table 3 takes the component-wise cost and determines the usage of each of the materials. The theoretical usage simply sums the cost of each of the components using the material. The actual usage takes the usable percentage of the material consumed; this is best visualized in Figure 6. This table constitutes the bill of materials with respect to the raw materials.

Material	Allotted	Theoretical Usage	Actual Usage
Balsa, 2 sheets $4 \times 36 \times 1/8$ in	288	211 (73%)	274 (95%)
Balsa, 2 sheets $4 \times 18 \times 1/8$ in	144	98.8 (69%)	112 (78%)
Spruce, 2 spars $3/8 \times 3/16 \times 36$ in	27	27 (100%)	27 (100%)
Tissue paper, 20×36 in	720	512 (71 %)	N/A

Table 3: Material Allotment and Usage (in^2)

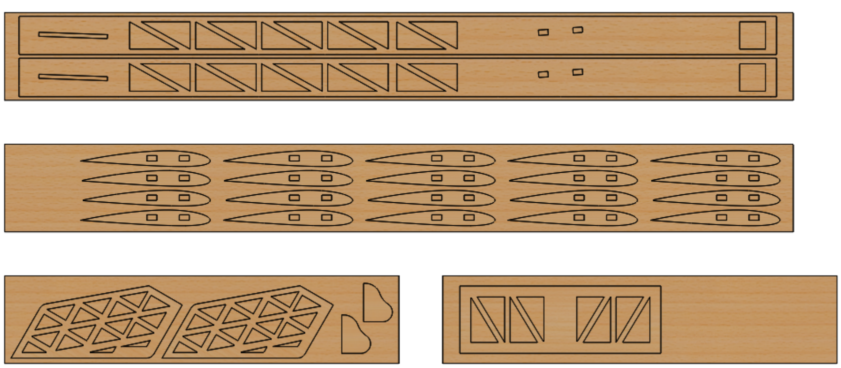

Figure 6: Outlines of laser cutting for balsa wood components

Once all the components are laser cut, the glider will be assembled by first setting the horizontal and vertical stabilizers and then the wings. Then, tissue paper will be used to create the wing's surface and to close the mass-reduction holes. Finally, ballast will be added with empirical testing to determine the center of mass and stability of the glider. Empirical testing before incorporating ballast is vital for ensuring that the mass of the tissue paper, glue, and other non-modeled components is properly considered in the glider's stability.

GLIDER COMPETITION REFLECTION
MAE 3050—Introduction to Aeronautics

Group 14: Nagamitesh Nagamuralee, Nicholas Schneider, Pranav Shankar
Date: December 13, 2025

1. *Competition Discussion:*
   The prediction for our flight time was around 5 seconds to travel 20 meters. In the competition our glider flew close to 10 meters. Its sink rate was much higher than anticipated and from the launch it did not retain much horizontal momentum, mostly due to the additional mass added in clay to move the COM forward. This was done to move the COM ahead of the wings, as it previously was behind it. We ended up adding close to 40g in order to achieve this. The glider flew steadily for the first few seconds then stalled not because its pitch was too high but due to a lack of velocity.

   After our first stall attempt our front nose broke off, and to rectify the stalling that we saw during the first run we attempted to use some clay to stick the nose back on and then we removed some clay as well to reduce the overall mass of the aircraft to reduce its sink rate. We also increased power in hopes that it would travel farther, and our second attempt did go a bit better and we did go a little further but not by much and still rapidly sank in a similar amount of time. We also noticed that our glider veered to one direction in the first run, so in this second run we added a little clay on the opposite wing to allow it to have a more steady and straight flight.

   Another minor change that we did for our flights is that we added a small notch in the tail section of our glider, the reason we needed this is that we had trouble keeping the elastic on the glider as we launched since our tail has an angle and that angle made the elastic slip underneath our glider and resulted in a false start before our first flight. The notch helped hold the elastic onto the plane until it fully launched and improved transfer of energy from the rubber band to the plane.

2. *Next Steps:*
   Using the same glider, I think that the biggest help would be to find ways to cut down weight on the back of our glider (maybe one piece of balsa in the back rather than two), one to save mass and secondly to push our center of mass forward which would allow us to use less clay to shift our center of mass. I think we would also characterize the rubber bands and friction on the launching surface more since each launching surface had a different stiffness of rubberband and imparted different forces it seemed like.

I think that the biggest thing that would have helped us if we start over is accounting for how much clay or mass we would need to move the center of mass forward during the design phase. In this iteration even though all of our calculations were correct, we didn’t fully factor in the impact of overall mass caused by the clay and we were around 20 grams heavier than we wanted to be meaning that our glider sank really fast since there wasn’t enough lift to counteract the gravity force pulling the plane down. I would also try to avoid using clay and build the glider differently, maybe with a longer front section such that we don’t even need to use clay.